

Machine Learning (ML)

Mid-Semester Exam Preparation — Unit I & Unit II
Easy English Definitions + Hinglish Explanation + Short Examples

Exam Tip: Learn the definition and headings first. In long answers, add explanation, example, diagram/flow, and a short conclusion.

UNIT I

Q1. Define Machine Learning. Write the Objectives of Machine Learning.

Easy English Definition: Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

Hinglish Explanation: ML mein computer ko har rule manually nahi batana padta. Hum data/examples dete hain, model patterns learn karta hai, aur new data par prediction ya decision leta hai.

Objectives:

- **Prediction:** Predict future outcomes from past data.
- **Classification:** Divide data into predefined classes, e.g., Spam/Not Spam.
- **Pattern Recognition:** Find useful patterns in data.
- **Decision Making:** Support or automate decisions using learned patterns.
- **Automation:** Automate repetitive tasks.
- **Learning and Improvement:** Improve performance using relevant data and experience.
- **Knowledge Discovery:** Discover useful or hidden information from large datasets.

Short Example: A spam detector learns from previous emails and predicts whether a new email is spam or not.

Remember: Learn from Data → Find Patterns → Predict/Decide

Q2. Explain Different Learning Models. Compare Supervised and Unsupervised Learning Methods.

Easy English Definition: A learning model describes how a machine learning system learns from available data.

Hinglish Explanation: Learning model ka matlab hai machine ko kis type ke data se aur kis tarike se learn karaya ja raha hai.

Major Learning Models:

- **Supervised Learning:** Model learns from labelled data, where input and correct output are known.
- **Unsupervised Learning:** Model learns from unlabelled data and finds hidden groups or patterns.
- **Semi-Supervised Learning:** Uses a small amount of labelled data together with a large amount of unlabelled data.
- **Reinforcement Learning:** An agent learns by interacting with an environment and receiving rewards or penalties.

Supervised vs Unsupervised:

Supervised: Labelled data → learns input-output relationship → common tasks: classification and regression.

Unsupervised: Unlabelled data → discovers structure/patterns → common tasks: clustering and association.

Short Example: Supervised: emails labelled spam/not spam. Unsupervised: group customers into similar behaviour groups without predefined labels.

Q3. Write the Name and Explain Each Process of KDD in Machine Learning.

Easy English Definition: KDD (Knowledge Discovery in Databases) is the process of finding useful and meaningful knowledge from large datasets.

Hinglish Explanation: KDD ek complete process hai jisme raw data ko select, clean, transform aur analyse karke useful knowledge nikala jata hai.

KDD Steps:

- **1. Data Selection:** Select the required data from available data sources.
- **2. Data Cleaning / Preprocessing:** Remove or handle missing, noisy, inconsistent, or duplicate data.
- **3. Data Transformation:** Convert data into a suitable format for analysis.
- **4. Data Mining:** Apply algorithms to find patterns or relationships.
- **5. Pattern Evaluation:** Identify patterns that are useful, valid, and meaningful.
- **6. Knowledge Presentation:** Present the discovered knowledge using reports, rules, charts, or visualization.

KDD Flow: Data Selection → Cleaning → Transformation → Data Mining → Pattern Evaluation → Knowledge Presentation

Short Example: A supermarket selects sales data, cleans it, transforms it, mines buying patterns, evaluates useful patterns, and presents “customers who buy X often buy Y.”

Q4. What is SEMMA? Explain in Detail with Example.

Easy English Definition: SEMMA is a data mining methodology developed by SAS. It stands for Sample, Explore, Modify, Model, and Assess.

Hinglish Explanation: SEMMA ek step-by-step approach hai jo data ko sample karne se lekar model ko assess karne tak data mining process ko organize karta hai.

SEMMA Steps:

- **Sample:** Select a representative sample of data for analysis.
- **Explore:** Study the data to find patterns, relationships, trends, and unusual values.
- **Modify:** Clean, transform, select, or create useful variables/features.
- **Model:** Build a predictive or descriptive model using suitable algorithms.
- **Assess:** Evaluate the model and check whether it gives useful and reliable results.

Short Example: For customer churn prediction: Sample customer records → Explore churn patterns → Modify features → Model churn → Assess model performance.

Remember: S-E-M-M-A = Sample → Explore → Modify → Model → Assess

Q5. Write About the History and Evolution of Machine Learning.

Easy English Definition: The history of Machine Learning describes how computers evolved from rule-based programs to systems that learn from data and support modern AI applications.

Hinglish Explanation: ML ekdum se nahi bana. Pehle computers fixed rules follow karte the, phir statistical learning aayi, neural networks improve hue, aur large data + computing power ke saath modern ML/deep learning develop hua.

Major Stages:

- **1950s–1960s:** Early AI and learning ideas developed; simple learning programs and neural network research appeared.
- **1970s–1980s:** Expert systems and knowledge-based approaches became important; statistical methods also developed.
- **1990s:** Data-driven machine learning became more practical, with methods such as decision trees, SVMs, and probabilistic techniques.
- **2000s:** Growth of internet data, better hardware, and larger datasets increased the use of machine learning.
- **2010s:** Deep learning achieved major improvements in image, speech, and language tasks.
- **2020s:** Generative AI, large language models, foundation models, and advanced AI applications became prominent.

Short Example: A system can evolve from fixed “if-then” rules to a model that learns patterns from millions of examples.

UNIT II

Q5. Explain ROC Curve with Diagram.

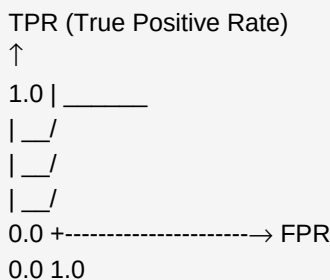
Easy English Definition: ROC (Receiver Operating Characteristic) Curve is a graph used to evaluate a binary classification model at different decision thresholds.

Hinglish Explanation: Classifier ka threshold change karne par True Positive Rate aur False Positive Rate change hote hain. ROC curve in dono ko graph par show karti hai.

Axes:

- **X-axis:** False Positive Rate (FPR) — proportion of actual negative cases incorrectly predicted as positive.
- **Y-axis:** True Positive Rate (TPR), also called Recall/Sensitivity — proportion of actual positive cases correctly predicted as positive.
- **Diagonal Line:** Represents random/no-skill classification.
- **Better Curve:** A curve closer to the top-left generally indicates better classification performance.
- **AUC:** Area Under the ROC Curve; it summarizes the model's ability to distinguish positive and negative classes. Higher AUC is generally better.

Conceptual Diagram:



Short Example: A disease classifier can use different thresholds. ROC shows how sensitivity changes as the false-positive rate changes.

Q6. Explain Polynomial Regression and Logistic Regression.

A. Polynomial Regression

Easy English Definition: Polynomial Regression is a regression method that models the relationship between input and output using polynomial terms, allowing a curved relationship.

Hinglish Explanation: Linear regression straight-line relationship model karta hai. Agar data curved pattern follow kare, toh polynomial regression x^2 , x^3 jaise terms use karke curve fit kar sakta hai.

General Form: $y = b_0 + b_1x + b_2x^2 + \dots + b_nx^n$

Example: Predicting crop growth when growth increases and then slows down can be represented by a curve.

B. Logistic Regression

Easy English Definition: Logistic Regression is a classification algorithm used mainly for binary outcomes, where the output is a probability between 0 and 1.

Hinglish Explanation: Logistic regression mainly classification ke liye use hota hai. Ye probability calculate karta hai aur threshold ke basis par class decide hoti hai.

Example: Predict whether an email is Spam or Not Spam, or whether a customer will churn or not.

Key Difference: Polynomial Regression → predicts a continuous numerical value and can model curves. Logistic Regression → predicts class probability and is mainly used for classification.

Q7. Explain Scales of Measurement.

Easy English Definition: Scales of measurement describe how variables are measured and what mathematical operations are meaningful for them.

Hinglish Explanation: Measurement scale batata hai ki kisi variable ka data kis type ka hai aur us data par hum kaunsi calculations meaningfully kar sakte hain.

Four Scales:

- **1. Nominal:** Categories with names/labels and no natural order. Example: Gender category, city, blood group.
- **2. Ordinal:** Categories with a meaningful order, but differences between levels are not necessarily equal. Example: Low, Medium, High.
- **3. Interval:** Ordered numerical values with equal intervals, but zero is not a true absence. Example: Temperature in Celsius.
- **4. Ratio:** Numerical values with equal intervals and a meaningful zero. Example: Age, weight, height, income.

Memory Trick: Nominal = Name | Ordinal = Order | Interval = Equal Interval | Ratio = Real Zero

Q8(i). Short Note: Feature Construction or Generation

Easy English Definition: Feature Construction or Generation is the process of creating new useful features from existing data to help a machine learning model learn better.

Hinglish Explanation: Existing columns ko use karke ek naya meaningful feature banana feature construction/generation hai.

Examples:

- Date of birth → Age
- Price and Quantity → Total Amount
- Height and Weight → BMI
- First Name + Last Name → Full Name

Why Useful? Better features can make patterns easier for the model to learn and may improve model performance.

Q8(ii). Short Note: Correlation and Causation

Easy English Definition: Correlation means two variables change together, while causation means a change in one variable directly causes a change in another.

Hinglish Explanation: Correlation ka matlab relation/association hai; causation ka matlab actual cause-effect relationship. Correlation alone does not prove causation.

Example: Ice-cream sales and swimming activity may both increase in summer. They are correlated, but ice-cream sales do not necessarily cause swimming; a third factor, such as hot weather, can affect both.

Remember: Correlation $\neq$ Causation

Q9. Explain Perspective of Data in Machine Learning.

Easy English Definition: The data perspective in Machine Learning focuses on understanding data as the main input from which a model learns patterns, relationships, and useful information.

Hinglish Explanation: ML ke liye data sabse important input hai. Data ko uske type, structure, quality, features, labels, and relationships ke perspective se samajhna zaroori hai.

Main Points:

- **Data as Input:** Models learn from available data.
- **Features:** Input variables used to describe examples.
- **Labels/Target:** The desired output in supervised learning.
- **Data Quality:** Missing, noisy, duplicate, or incorrect data can affect learning.
- **Data Types:** Numerical, categorical, text, image, time-series, etc.

- **Data Distribution:** Understanding how values are distributed helps choose suitable methods.
- **Training and Testing:** Data is commonly divided so model performance can be evaluated on unseen data.
- **High-Dimensional Data:** Many features can make learning difficult and may require feature selection or dimensionality reduction.

Short Example: For house-price prediction, area, number of rooms, location, and age are features; house price is the target.

Q10. Elaborate Scales of Measurement with Respect to Machine Learning.

Easy English Definition: In Machine Learning, scales of measurement help us understand the type and properties of each variable so that suitable preprocessing, analysis, and algorithms can be selected.

Hinglish Explanation: ML mein variable ka measurement scale pata hona important hai because har data type par same mathematical operation nahi kar sakte.

1. Nominal Scale

Used for labels/categories without order. Example: Color = Red, Blue, Green. Usually represented using methods such as one-hot encoding when an ML algorithm needs numeric input.

2. Ordinal Scale

Categories have order. Example: Satisfaction = Low, Medium, High. The order is meaningful, but the gap between categories should not automatically be treated as equal.

3. Interval Scale

Values have equal intervals, but zero is not a true absence. Example: Celsius temperature. Addition/subtraction can be meaningful, but ratio statements such as "20°C is twice 10°C" are not valid.

4. Ratio Scale

Values have equal intervals and a meaningful zero. Example: age, weight, height, income. Ratio comparisons are meaningful.

Why It Matters in ML:

- Helps choose suitable preprocessing and encoding methods.
- Prevents incorrect mathematical interpretation of categorical data.
- Helps decide which summary statistics are meaningful.
- Supports better feature engineering and model selection.
- Makes data interpretation more reliable.

Comparison:

Nominal: Name/no order | **Ordinal:** Order/no guaranteed equal gaps | **Interval:** Equal gaps/no true zero | **Ratio:** Equal gaps + true zero

■ LAST-MINUTE REVISION SHEET

Unit I: ML Definition/Objectives → Learning Models → KDD → SEMMA → History & Evolution

Unit II: ROC → Polynomial & Logistic Regression → Measurement Scales → Feature Construction → Correlation vs Causation → Data Perspective

Must-Remember: KDD = Selection → Cleaning → Transformation → Data Mining → Pattern Evaluation → Knowledge Presentation

SEMMA = Sample → Explore → Modify → Model → Assess

Scales = Nominal → Ordinal → Interval → Ratio

ROC = TPR vs FPR; AUC summarizes discrimination ability.

Exam Strategy: For long answers: Definition → headings/points → explanation → short example → diagram/flow if useful → conclusion.